

Real-Time Market Movement Prediction System: A Machine Learning and MLOps Approach

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Abstract—This project presents a real-time market movement prediction system using machine learning and deep learning techniques integrated with MLOps practices. The system combines historical stock market data and financial news sentiment to predict directional price movements. Recurrent neural networks including RNN, LSTM, and GRU are implemented and evaluated. The system is automated using Apache Airflow and version-controlled using DVC to ensure reproducibility.

Index Terms—Stock Prediction, LSTM, RNN, GRU, Time-Series Forecasting, MLOps, Airflow, DVC

I. INTRODUCTION

Financial markets are highly dynamic and influenced by multiple unpredictable factors. Traditional statistical models fail to capture complex temporal dependencies in financial data. This project addresses these limitations using machine learning and deep learning techniques for real-time market prediction.

II. METHODOLOGY

A. Data Collection Process

Market data was collected using automated scripts (`data_ingestion.py`) from multiple stock tickers including AAPL, TSLA, MSFT, GOOGL, and AMZN. Financial news was collected via RSS feeds and stored in a structured database.

B. Preprocessing Steps

The dataset was processed using:

- Missing value handling
- Min-Max normalization
- Time-series sequence generation
- Sentiment aggregation from news articles

C. Model Architecture Explanation

The system evaluates three recurrent architectures:

- RNN (Simple recurrent model)
- LSTM (Long Short-Term Memory)
- GRU (Gated Recurrent Unit)

LSTM and GRU were expected to perform better due to gating mechanisms; however, performance varied due to dataset size and noise.

D. Reasoning Behind Model Selection

RNN was selected for baseline comparison, while LSTM and GRU were chosen for their ability to capture long-term dependencies in sequential data.

III. DATASET OVERVIEW

A. Data Sources

- Historical stock prices (AAPL, TSLA, MSFT, GOOGL, AMZN)
- Financial news RSS feeds
- Sentiment analysis outputs

B. Dataset Structure

The dataset is structured as time-series sequences:

$$X \in R^{270 \times 5 \times 9}$$

C. Class Distribution

The dataset is relatively balanced:

- 53% UP movement
- 47% DOWN movement

D. Challenges

- Noisy financial data
- Missing timestamps in news feeds
- Data imbalance across time windows

IV. SYSTEM ARCHITECTURE

A. Data Pipeline Architecture

Includes ingestion from APIs and RSS feeds, followed by preprocessing and feature engineering.

B. Training Pipeline

Includes sequence generation, model training (RNN/LSTM/GRU), and evaluation.

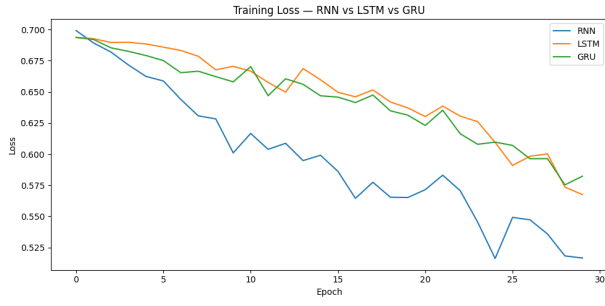


Fig. 1. End-to-End System Architecture

C. Inference System

Real-time predictions are generated through a Flask-based dashboard.

V. EXPERIMENTS AND RESULTS

A. Model Comparison

TABLE I
PERFORMANCE COMPARISON

Model	Accuracy	F1-Score
RNN	48.1%	0.588
LSTM	44.4%	0.516
GRU	42.6%	0.523

B. Evaluation Discussion

RNN performed slightly better than LSTM and GRU due to dataset size limitations and high noise levels. Deep models require larger datasets to fully utilize their learning capacity.

C. MLflow Usage

MLflow was not used due to project scope limitations. Model tracking was handled manually through logs and saved result files.

VI. PROBLEMS FACED

- Noisy and inconsistent financial data
- Overfitting in deep learning models
- Limited dataset size
- Real-time data synchronization issues
- High computational cost during training

VII. FINDINGS AND DISCUSSION

The results indicate that model performance depends heavily on dataset size and quality. Simpler models such as RNN can outperform deeper architectures when data is limited. Sentiment analysis improves feature richness but introduces noise if not weighted properly.

VIII. CONCLUSION AND FUTURE WORK

This project successfully developed a real-time market prediction system using machine learning and MLOps tools. Future improvements include transformer-based models, larger datasets, and cloud-based deployment.

IX. REFERENCES

REFERENCES

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